

## Video Sharing

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# Technical Description

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## Change History

|                   |      |                          |
|-------------------|------|--------------------------|
| February 11, 2005 | V1.0 | Initial document release |
|                   |      |                          |

# 1 Introduction

Video sharing enriches voice calls through sharing live or prerecorded video clips between peer-to-peer devices, based on standards of the Third-Generation Partnership Project (3GPP) and Internet Engineering Task Force (IETF). The purpose of this document is to specify the technical details and recommendations that must be followed in order to achieve interoperability between a Nokia video-sharing application and any other application capable of sending and receiving video during a voice call.

The primary user device running a video sharing application is a mobile phone operating over a cellular network or WLAN. The radio access to the mobile network may be GERAN (EGPRS with DTM), UTRAN, or CDMA2000. Video sharing applications use services of the IP Multimedia Subsystem (IMS) as described in the 3GPP and 3GPP2 specifications or Session Initiation Protocol (SIP) elements called proxies (see References 1 and 2). Hence, a video sharing application may also be running on a device attached to a fixed IP network (the Internet or a corporate intranet) and the recipient device may be a mobile phone, PC, or other device supporting IP connections.

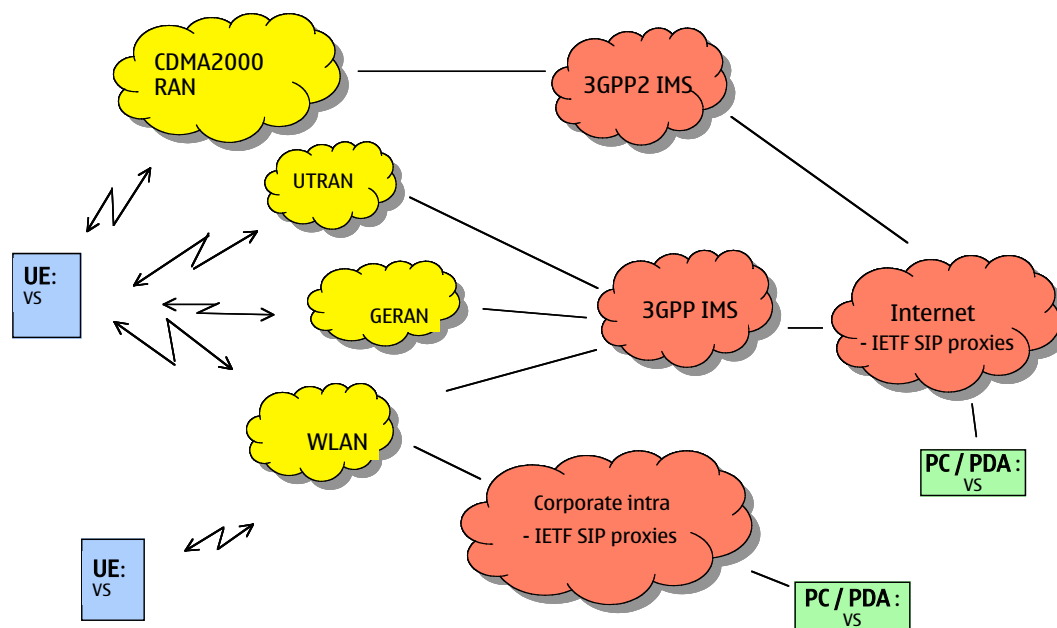


Figure 1 Video sharing network architecture  
Mobile User Equipment (UE) has a number of network technologies available for video sharing.

The primary use case for video sharing is to enrich a voice call through the sharing of live video or a prerecorded video clip. Therefore, simultaneous circuit switched (CS) and packet switched (PS) connections for voice and video are needed, and this is achieved in GERAN networks with DTM. WCDMA networks inherently support simultaneous CS and PS connections.

Chapter 2 presents the audio/video format used by the video sharing application. Chapter 3 presents diagrams to clarify end-to-end transfers and the propagation delays of media. Chapter 4 presents the usage of SIP and the Session Definition Protocol (SDP). Chapter 5 presents the usage of the Real-time Transport Protocol (RTP) and the RTP Control Protocol (RTCP). Chapter 6 takes a short look at usage of IP protocols by a video sharing application. Chapter 7 addresses the video sharing application and its use of various radio-access alternatives.

## 2 Video Sharing And Media Encoding

Video sharing uses the same audio and video codecs that are mandatory for 3GPP Packet Switched Streaming (PSS) and conversational multimedia applications.

### 2.1 Video

As a baseline, a video sharing application supports the following video parameters:

- ◆ Video Codec: H.263 Profile 0 Level 10.
- ◆ Bit rate: 57 kbps.
- ◆ Frame rate: 15 fps.
- ◆ Frame size: QCIF.
- ◆ Payload size: 500 bytes (not constant).
- ◆ RTP packetization.

A video sharing application supports H.263 Profile 0 Level 10 (and Level 45) video (H263-2000 RTP payload format). It may also support other video codecs such as H.264/AVC baseline and MPEG4 Visual Simple Profile 0 Level 0.

Recommended video (H.263 Profile Level 10) bit rate of a video sharing application is 57 kbps<sup>1</sup> with QCIF frame size and 15 fps frame rate. However, because of the network load, a lower bit rate may have to be applied; 20 kbps at 5 fps can be considered as the minimum video bit rate. Higher frame rates and frame sizes with the same or other codecs will be supported in the future. A video sharing application will typically include 500 bytes of video data in one RTP packet, but this is not constant. To save bandwidth or because of encoding issues, the payload size may be increased up to about 1,000 bytes.

### 2.2 Audio

The sending video-sharing application must provide video, but audio is optional. As a baseline, the audio stream has the following parameters:

- ◆ Audio Codec: AMR NB.
- ◆ Codec Mode: 12.2 kbps.
- ◆ RTP payload format: bandwidth efficient.
- ◆ Number of AMR speech frames in RTP packet: 20.

The video sharing application supports the AMR NB audio codec. The goal is to use the 12.2-kbps mode for good speech quality, but in conditions with limited network resources the AMR NB bit rate can be decreased down to the lowest bit rate, that is 4.75 kbps. The RTP payload format used is bandwidth efficient. The 20 speech frames in each RTP packet would result in a small protocol overhead, but the number is implementation dependent and can be thought of as a recommendation for the upper limit. Using the 12.2-kbps audio bit rate with 20 speech frames in each RTP packet would require a total of

<sup>1</sup> It is assumed that a 64-kbps WCDMA radio bearer is available for video transmission. Because of typical 60 bytes RTP/UDP/IPv6 protocol overhead, the video bit rate of 57 kbps with 500 bytes payload takes a total of about 64 kbps. Note that it is up to the radio network to decide whether to use header compression while applications should apply bit rates without it.

about 13 kbps of bandwidth. Considering audio taking 13 kbps from a 64-kbps radio bearer, video would have 51 kbps with protocol headers leading to about a 45-kbps video bit rate.

### 3 Media Delay

The goal is for the video provided by the sending device to be presented in the receiving device with minimal delay. In the near future, a good target would be for an end-to-end delay for media of one to two seconds, but it is to be noted that the network delay can vary a lot depending on the network load.

The networks cannot prevent some small jitter for the media, but still there should not be interruptions in the presentation of the video in the receiving UE. This requirement creates a need for buffering in the receiving UE. The receiving device should be capable of adjusting its buffering based on analysis of RTCP reports or an SDP description indicating the QoS capability (see chapter 4). See chapter 7 for examples of target delays in different network environments.



## 4 Usage Of SIP And SDP

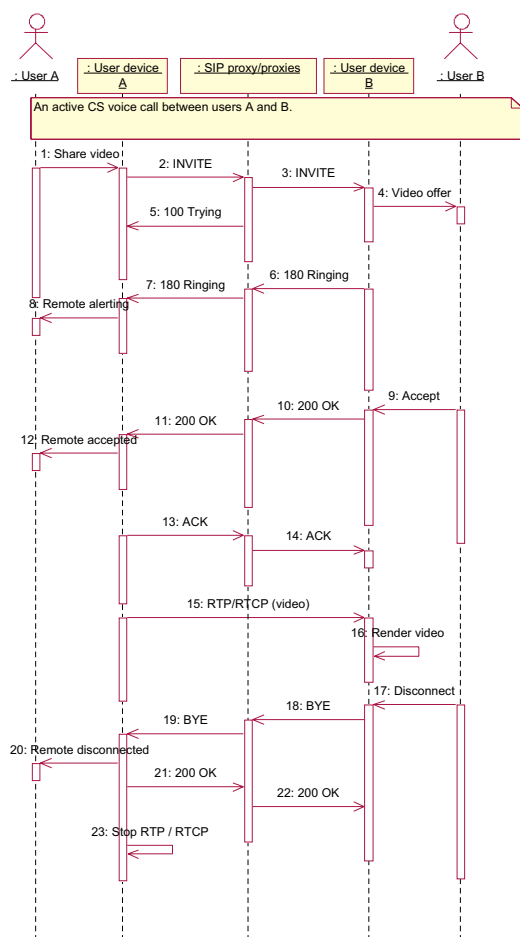


Figure 2 IETF SIP signaling of a video sharing session

Video sharing uses SIP and SDP as described for the IETF, 3GPP IMS, and 3GPP2 IMS domains. SIP signaling in the IETF domain is simpler, as the 3GPP and 3GPP2 domains mandate additional SIP and SDP features.

Figure 2 presents the simplest IETF SIP signaling. In the primary video-sharing use case, an active CS voice call exists prior to the users' initiating the video sharing. PC clients on the Internet typically use this kind of signaling as well. Mobile devices apply this signaling if they are not connected to the SIP services of IMS (by default they are) but use a SIP proxy or proxies for Internet connections. Some SIP methods contain an SDP description (see Reference 2)

The following is an example of an SDP description in the media offer of a sending UE.

```

v=0
o=- 2435697 2435697 IN IP4 172.21.137.44
s=-
  
```

```

c=IN IP4 172.21.137.44
t=0 0
a=application:com.nokia.rtvS
a=sendonly
m=video 49152 RTP/AVP 96
a=rtpmap:96 H263-2000/90000m=audio 57344 RTP/AVP 97a=rtpmap:97 AMR/8000T

```

In this example, audio is included in the media offer, which would be the case if video is streamed from a file. This kind of SDP description could be included in SIP signaling when IMS is not used for SIP services (see References 8 and 16 for details).

The sending UE indicates unidirectional media stream with *sendonly* attribute. *a=application:com.nokia.rtvS* identifies the application (may be used by the network for statistical purposes).

The following is an example of the SDP description in the media answer of the receiving UE:

```

v=0
o=- 308519342 2 IN IP4 172.168.1.55
s=-
c=IN IP4 172.168.1.55
t=0 0
a=application:com.nokia.rtvS
a=recvonly
m=video 49152 RTP/AVP 96
b=AS:57
a=rtpmap:96 H263-2000/90000
a=fmtp:96 PROFILE=0;LEVEL=10
m=audio 57344 RTP/AVP 97
b=AS:8
a=rtpmap:97 AMR/8000
a=maxptime:500

```

The receiving UE would use *a=recvonly* attribute to indicate that it will only receive. *b=AS:57* indicates the maximum video bit rate in one direction. *a=fmtp:96 profile=0;level=10* indicates support for profile 0 and level 10 by the receiving UE. *b=AS:8* indicates the audio bit rate including RTP/UDP/IP headers and RTCP packets when AMR mode 0 is used. *a=maxptime* indicates that the maximum supported packet size is 500 milliseconds.

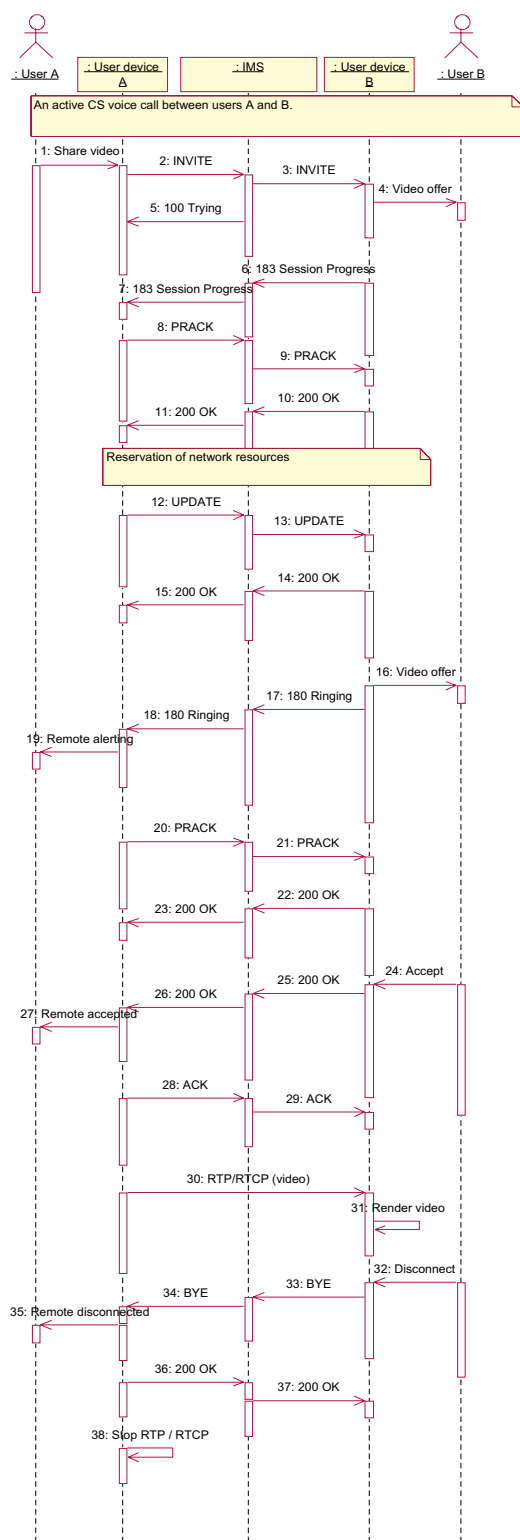


Figure 3 3GPP SIP signaling of a video sharing session

Figure 3 presents the SIP signaling that is applied by mobile devices when using the SIP services of IMS. Some SIP methods contain an SDP description (see Reference 6). The following is an example of a 3GPP SDP description in the media offer of a sending UE:

```
v=0

o=- 2987933615 2987933615 IN IP6 5555::aaa:bbb:ccc:ddd

s=-

c=IN IP6 5555::aaa:bbb:ccc:ddd

t=0 0

a=application:com.nokia.rtvS
a=sendonly

m=video 49152 RTP/AVP 96

a=rtpmap:96 H263-2000/90000

a=curr:qos local none

a=curr:qos remote none

a=des:qos mandatory local send

a=des:qos none remote sendm=audio 57344 RTP/AVP 97

a=rtpmap:97 AMR/8000
```

The *a=curr:qos* and *a=des:qos* attributes are related to QoS preconditions that are mandatory in 3GPP Rel5 (see Reference 17 for details). The following is an example of the SDP description in the media answer of the receiving UE:

```
v=0

o=- 2987933623 2987933623 IN IP6 5555::eee:fff:aaa:bbb

s=-

c=IN IP6 5555::eee:fff:aaa:bbb

t=0 0

a=application:com.nokia.rtvS
a=recvonly

m=video 49152 RTP/AVP 96

a=rtpmap:96 H263-2000/90000

a=fmtp:96 PROFILE=0;LEVEL=10

b=AS:57
```

```

a=curr:qos local none
a=curr:qos remote none
a=des:qos mandatory remote recv
a=des:qos mandatory local recv
a=conf:qos remote recv
m=audio 57344 RTP/AVP 97
b=AS:8
a=curr:qos local none
a=curr:qos remote none
a=des:qos mandatory remote recv
a=des:qos mandatory local recv
a=conf:qos remote recv
a=rtpmap:97 AMR/8000
a=maxptime:500

```

The attribute *a=curr:qos* is related to QoS preconditions that are mandatory in 3GPP; *a=conf:qos remote recv* indicates a willingness to get confirmation of resource reservation. See Reference 17 for more details.

Some of the media parameters are optional in an SDP description. The video sharing application assumes the following default values if the related parameters are not included in SDP:

- ◆ Video bit rate, default: 57 kbps.
- ◆ Video frame rate, default: 15 fps.
- ◆ Video resolution, default QCIF.
- ◆ AMR packetization, default: 400 milliseconds (20 speech frames) in each RTP packet (audio is optional).

## 5 Usage Of RTP And RTCP

Video and optionally audio (for example, if source of video is a file with audio) are sent in RTP packets. See sections 2.1 and 2.2 for RTP packetization-related information.

The sending video-sharing application sends RTCP sender reports and the receiving video-sharing application sends RTCP receiver reports. The interval for sending the RTCP reports is an implementation issue depending on, for example, bearer capabilities. First implementations use interval of three seconds for sender and receiver reports, but the interval may change in the future.

### 5.1 RTP And RTCP Ports

Source and destination ports are set in the UDP header. By default, the video sharing application uses the following source ports: 49152 for video, 49153 for video-related RTCP, 57344 for audio, and 57345 for audio-related RTCP. The specific port numbers are used to provide control and security to the mobile network administrator.

### 5.2 Symmetric RTP

The terms symmetric RTP and RTCP are defined in Reference 26. Operators protect their packet networks with firewalls, which typically allow symmetric RTP and RTCP traffic. Because of the unidirectional video of video sharing applications, the UE receiving video has to “simulate” symmetric RTP traffic by sending an RTP packet uplink. Usage of symmetric RTP gives security to video sharing sessions between UEs in different operators’ networks.

Video sharing applications use symmetric RTP in the following way: Immediately after the UE accepting the video offer sends SIP 200 OK method for INVITE, it starts sending once every 30 seconds a “dummy” RTP packet (one for video, another one for audio if used) to the sending UE. The “dummy” RTP packets have no payload but the payload type in the RTP header indicates the RTP stream it is related to by using the payload type of audio or video RTP packets. SSRC of “dummy” RTP packets is different from those of RTP packets carrying media and remains the same during one video-sharing session. SN is increased as for any RTP stream while TS can be anything. The source port in the UDP header is set to the value that is used for receiving media and destination port to the value that is used as source port by the sending UE. Symmetric RTP should be disabled if the UE and network elements support service-based local policy as presented in Reference 1.

### 5.3 Rate Adaptation

Applications must be able to adapt to changing network conditions. The sending UE adjusts the media details to the changing network conditions using RTCP receiver reports and knowledge about the available network. The receiving UE must be able to adapt to changing bit rates within the limits negotiated by SIP or SDP.

## 6 Internet Protocols

Different network domains, as described in Chapter 2, support either IPv4 or IPv6, and the video sharing application may also use either of them. RTP/UDP/IPv4 presents 42 bytes protocol overhead for each sent packet while RTP/UDP/IPv6 presents typically 60 bytes overhead. Interworking of network domains with different IP versions is out of the scope of this document.

## 7 Video Sharing And Various Radio Accesses

### 7.1 Video Sharing In 3GPP Networks

#### 7.1.1 EGPRS and WCDMA radio

Multislot classes (see Reference 10) define how many timeslots in uplink and downlink direction are used to carry data over the EGPRS radio connection. Terminals must be able to apply multislot classes with enough timeslots in order to provide appropriate bandwidth to transmit video. Using CS speech in parallel with PS video, which is enabled by Dual Transfer Mode (DTM), takes at least half of a timeslot (depending on the configuration) in addition to the timeslots occupied by PS video.

In WCDMA, networks radio bearers (RB) and radio access bearers (RAB), rather than timeslots and multislot classes, will define the capacity of the radio interface. Reference 9 lists examples of RABs and RBs supported by UMTS Terrestrial Radio Access (UTRA). Sections 7.1.95 and 7.1.96 of that document are examples of RAB combinations applicable to video sharing. In those examples, “streaming / unknown / DL (or UL) 128 kbps / PS RAB” is the one that could carry the media of video sharing. This is just one example, however, and the bandwidth could also be 64 kbps, and the traffic class could be interactive or background depending on the type of the Packet Data Protocol (PDP) context (see sections 7.1.3 and 7.1.4).

#### 7.1.2 PDP contexts

The PDP context is an abstraction of the PS connection in GERAN and UTRAN networks. Sections 7.1.3 and 7.1.4 give examples of sets of 3GPP (Rel99) QoS parameters for PDP contexts, assuming that only video is sent.

Video sharing has the following order of preference for the traffic class of the media PDP context: 1) streaming and 2) interactive/background. The network may downgrade the requested traffic class and other parameters and the video sharing application must be able to adjust. So while it is preferred to aim at the QoS parameters in Table 2, the video sharing application must be able to adjust different parameters, for example, to those of Table 1. In fact, the parameters presented in Table 1 may be reality in the near future.

#### 7.1.3 Video sharing over best-effort connections

A best-effort connection is used here to refer to a network connection that does not have guarantees for the available bandwidth or introduced delays. Table 1 gives an example of a Rel99 QoS parameter set that could be applied for the PDP context used to deliver video sharing media over a best-effort connection. The parameters are for a sending UE. A receiving UE would have otherwise the same parameters except that the uplink and downlink bit rates are interchanged. Best-effort connections can be expected to be common in the near future.



| Traffic class                    | Interactive                              |
|----------------------------------|------------------------------------------|
| Traffic handling priority        | 1                                        |
| Delivery order                   | No                                       |
| Maximum SDU size                 | 1,200 bytes                              |
| Delivery of erroneous SDUs       | No                                       |
| SDU error ratio                  | 10e-4                                    |
| Residual BER                     | 10e-5                                    |
| Transfer delay                   | N/A                                      |
| Maximum bit rate for uplink      | Product specific maximum, e.g., 128 kbps |
| Guaranteed bit rate for uplink   | N/A                                      |
| Maximum bit rate for downlink    | Product specific maximum, e.g., 128 kbps |
| Guaranteed bit rate for downlink | N/A                                      |
| Source statistic descriptor      | Unknown                                  |

Table 1 Example of (UL) QoS parameters of an interactive PDP context, video only

Transfer delay and guaranteed bit rates are not defined for interactive traffic class. Background traffic class has the same QoS parameters except that traffic handling priority is not defined. Delay is heavily dependent on the network load and can be even a couple of seconds. The 4-kbps bit rate for downlink is used by RTCP reports.

#### 7.1.4 Video sharing over streaming connections

Streaming connection is used here to refer to a network connection that supports guaranteed bit rates and considerably low delays. As EGPRS and WCDMA networks proliferate, high-quality connections will be more dependable.

Table 2 presents an example of QoS parameters of a PDP context of streaming traffic class that has transfer delay and guaranteed bit rates defined. A delay of one second is a suitable value.

| Traffic class                    | Streaming                                |
|----------------------------------|------------------------------------------|
| Traffic handling priority        | N/A                                      |
| Delivery order                   | No                                       |
| Maximum SDU size                 | 560 Bytes                                |
| Delivery of erroneous SDUs       | No                                       |
| SDU error ratio                  | 10e-4                                    |
| Residual BER                     | 10e-5                                    |
| Transfer delay                   | 1,000 ms                                 |
| Maximum bit rate for uplink      | Product specific maximum, e.g., 128 kbps |
| Guaranteed bit rate for uplink   | 64 kbps                                  |
| Maximum bit rate for downlink    | Product specific maximum, e.g., 128 kbps |
| Guaranteed bit rate for downlink | 4 kbps                                   |
| Source statistic descriptor      | Unknown                                  |

Table 2 Example of (UL) QoS parameters of a streaming PDP context, video only

## 7.2 Video Sharing In 3GPP2 Networks

*To be provided in later versions.*

## 7.3 Video Sharing Over WLAN

WLAN offers attractive data speed, with a typical maximum bandwidth of 2 Mbps. This is remarkably higher than that needed by video sharing applications. However, WLAN operates in an unlicensed band and was not originally designed to guarantee exact QoS requirements. With the advent of real-time services, a need to improve the capacity of access points for real-time traffic has emerged. The WLAN QoS concept that is being defined in IEEE 802.11e (Reference 12) focuses on an air interface between WLAN access points and terminals, and the specification is still evolving. It defines a collection of Medium Access Control (MAC) protocol refinements, which makes it possible to prioritize and limit traffic in so-called access categories. The Wi-Fi forum focuses on defining its own test specifications for selected WLAN QoS features, based on subsets of IEEE 802.11e drafts, in order to accelerate the adoption of WLAN QoS enhancements in commercial implementations.

In order to maintain a constant service level for video sharing with low delay rates and high enough bit rate even in a congested WLAN network, video sharing implementations should apply WLAN QoS enhancements. However, current best-effort WLAN solutions can already provide a well-functioning video-sharing service in moderately crowded WLAN access networks.

## 8 Summary

Mobile network operators have shown great interest in video sharing. The technology builds on related 3GPP, IETF, and ITU-T specifications to implement the service between especially mobile devices and also between mobile and fixed terminals. Despite compliance with well-known standards, it is useful to note some details in order to achieve good interworking between Nokia video-sharing applications and other vendors' implementations. This document specifies the relevant details.

## 9 References

|    |                                                                                                                                                                             |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | 3GPP TS 23.228 v 5.12.0, IP Multimedia Subsystem                                                                                                                            |
| 2  | RFC 3261, SIP: Session Initiation Protocol                                                                                                                                  |
| 3  | Video Sharing: Enrich your voice call with video, published in Nokia Mobility Conference 2004                                                                               |
| 4  | 3GPP TS 26.071 v 5.0.0, AMR speech codec; general description                                                                                                               |
| 5  | ITU-T Recommendation H.263: Video coding for low bit rate communication                                                                                                     |
| 6  | 3GPP TS 24.229 v 5.9.0, IP Multimedia Call Control Based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP)                                        |
| 7  | RFC 2327 SDP: Session Description Protocol                                                                                                                                  |
| 8  | RFC 3264 An Offer/Answer Model with the Session Description Protocol (SDP)                                                                                                  |
| 9  | 3GPP TR 25.993 v 6.2.0                                                                                                                                                      |
| 10 | 3GPP TS 05.02 v 8.11.0 (Rel 99), TS 45.002 v 5.12.0 (Rel4)                                                                                                                  |
| 11 | IEEE 802.11e D5.0, Medium Access Control (MAC) Enhancements for Quality of Service (QoS), July 2003                                                                         |
| 12 | 3GPP2 P.S0001-A, Wireless IP Network Standard v3.0                                                                                                                          |
| 13 | RFC 3267 Real-Time Transport Protocol (RTP) Payload Format and File Storage Format for the Adaptive Multi-Rate (AMR) and Adaptive Multi-Rate Wideband (AMR-WB) Audio Codecs |
| 14 | RFC 3550 RTP: A Transport Protocol for Real-Time Applications                                                                                                               |
| 15 | RFC 3551 RTP Profile for Audio and Video Conferences with Minimal Control                                                                                                   |
| 16 | Draft-ietf-avt-rfc2429-bis-02 RTP Payload Format for the 1998 Version of ITU-T Rec. H.263 Video (H.263+)                                                                    |
| 17 | RFC 3312 Integration of Resource Management and Session Initiation Protocol (SIP)                                                                                           |
| 18 | RFC 3555 MIME Type Registration of RTP Payload Formats                                                                                                                      |
| 19 | 3GPP TS 26.235 v 5.1.0, Packet switched conversational multimedia applications; Default codecs                                                                              |
| 20 | 3GPP TS 26.236 v 5.5.0 Packet switched conversational multimedia applications; Transport protocols                                                                          |
| 21 | 3GPP2 X.P0013, IP Core Network — Multimedia Domain                                                                                                                          |
| 22 | 3GPP2 X.P0013.0, Multimedia Domain Overview                                                                                                                                 |
| 23 | 3GPP2 X.P0013.2, IP Multimedia Subsystem (IMS); Stage-2                                                                                                                     |
| 24 | 3GPP2 X.P0013.3, IP Multimedia (IM) session handling; IM call model                                                                                                         |
| 25 | 3GPP2 X.P0013.4, IP Multimedia Call Control Protocol based on SIP & SDP; Stage-3                                                                                            |
| 26 | draft-wing-mmusic-symmetric-rtprtcp-01, Symmetric RTP and RTCP Considered Helpful                                                                                           |
| 27 | ITU-T H.263 Annex X (03/2004), profiles and levels definition                                                                                                               |
| 28 | ITU-T Recommendation H.264                                                                                                                                                  |

## 10 Glossary

|          |                                       |
|----------|---------------------------------------|
| AP       | Access point                          |
| CDMA     | Code Division Multiple Access         |
| CDMA2000 | 3rd generation of CDMA                |
| CN       | Core network                          |
| CS       | Circuit Switched                      |
| DL       | Downlink                              |
| DTM      | Dual Transfer Mode                    |
| EDGE     | Enhanced Data Rates for GSM Evolution |
| EGPRS    | EDGE GPRS                             |
| GERAN    | GPRS Enhanced RAN                     |
| GPRS     | General Packet Radio Service          |
| LAN      | Local area network                    |
| NB       | Narrowband                            |
| PAN      | Personal area network                 |
| PoC      | Push-to-Talk over Cellular            |
| PS       | Packet switched                       |
| PSS      | Packet-switched streaming             |
| QoS      | Quality of Service                    |
| RAB      | Radio access bearer                   |
| RAN      | Radio access network                  |
| RB       | Radio bearer                          |
| RTCP     | RTP Control Protocol                  |
| RTP      | Real-time Transport Protocol          |
| SBLP     | Service-based Local Policy            |
| SDP      | Session Description Protocol          |
| SIP      | Session Initiation Protocol           |
| SN       | Sequence Number                       |
| SSRC     | Synchronization Source                |
| TS       | Time stamp                            |
| UE       | User equipment                        |
| UL       | Uplink                                |
| UMTS     | Universal Mobile Telephone System     |

|       |                                           |
|-------|-------------------------------------------|
| UTRA  | UMTS Terrestrial Radio Access             |
| WB    | Wideband                                  |
| WCDMA | Wideband Carrier Division Multiple Access |
| WLAN  | Wireless LAN                              |